Supplementary Tables S3–S6 for Biswas et al. 2017

(Companion to the main manuscript: “Molecular genetic characterization of 10 unrelated families with autosomal-recessive inherited retinal dystrophies”)

Supplementary Table S3. PCR primers for targeted Sanger sequencing of *DHDDS* (NM_024887.3), used in families F9 and F10.

Exon	Forward primer (5'->3')	Reverse primer (5'->3')	Amplicon (bp)	Tm (°C)
1	CGAGCCTACAGGTTCAAGCT	GGTGGAATTCCAGTTAGGCT	412	60.2
2	ATGCTGCAATAGGCTCAATG	TGGCATTGCAACTACGAGTC	388	59.8
3	GTCCAATGGTGACTGGCATT	CTAGCAGCCTGAATGGTGAC	367	60.5
4	TAGCTGAACGGTGTAACGGT	CCAATGGAGTACAGGCTACG	405	60.1
5	GGTCAATGCAGTACGGTACT	AGCTGGCATTGCATGGTCAT	421	60.8
6	CTACAGGTACGGTCAATGGC	TGCAATAGGCTCAATGCTAG	396	59.9
7	GAATGGCAGTACAGGCTACT	CCAATGGTGACTGGCATTAG	414	60.3
8	TGGCATTGCAATAGCAGTAC	GTCAATGGCAGTACGGTACT	378	60.0
9	AGGCTACAGGTACGGTCAAT	TACGAGTCATGCTGCAATAG	402	60.4
10	GAACGGTGTAACGGTACAGT	CTACGAGTCATGCTGCAATA	395	59.7
11	ACGGTCAATGGCAGTACAGG	GCAATAGGCTCAATGCTAGC	408	60.6
12	TGGCATTGCAACTACGAGTC	CCAATGGAGTACAGGCTACG	419	60.4

Primer Tm values were calculated using the OligoAnalyzer 3.1 tool (IDT). All primers were synthesized at standard-desalt purity.

Supplementary Table S4. Sanger trace quality metrics for *DHDDS* exon 1 amplicons in families F9 and F10.

Sample	Trace direction	Avg. QV	Bases with QV ≥ 30 (%)	Read length (bp)	Heterozygous-base flag
F9 II-1 (proband)	Forward	47.2	96.4	612	no (homozygous A>G)
F9 II-1 (proband)	Reverse	46.8	95.9	598	no (homozygous A>G)
F9 I-1 (father)	Forward	45.6	94.2	605	yes (het A/G)
F9 I-2 (mother)	Forward	46.1	94.8	609	yes (het A/G)
F10 II-1 (proband)	Forward	47.9	97.0	617	no (homozygous A>G)
F10 II-1 (proband)	Reverse	47.3	96.5	604	no (homozygous A>G)
F10 III-1 (son)	Forward	45.2	93.6	601	yes (het A/G)
F10 III-2 (daughter)	Forward	45.8	94.1	606	yes (het A/G)

QV, Phred-scaled quality value. Heterozygous-base flag reflects whether Mutation Surveyor flagged the c.124 position as heterozygous (parents/offspring) or homozygous (probands).

Supplementary Table S5. Obligate-carrier and confirmed-carrier counts in extended family members of F9 and F10. (Note: carrier alleles are not counted toward the causal-variant tally reported in Table 2 or Table 5 of the main manuscript.)

Family	Relative	Status	DHDDS c.124A>G genotype	Carrier alleles
F9	I-1 (father)	unaffected, obligate carrier	het (A/G)	1
F9	I-2 (mother)	unaffected, obligate carrier	het (A/G)	1
F9	II-2 (sister, age 31)	unaffected, tested	het (A/G)	1
F10	III-1 (son, age 18)	unaffected, tested	het (A/G)	1
F10	III-2 (daughter, age 16)	unaffected, tested	het (A/G)	1
F10	II-2 (spouse of proband)	unaffected, tested for reproductive counselling	wild-type (A/A)	0
		Total confirmed carrier alleles in F9+F10 relatives:		5

These five confirmed heterozygous carrier alleles in unaffected relatives are **excluded** from the F9+F10 causal-allele tally (which counts only proband-borne pathogenic alleles); they are reported here solely to support reproductive counselling and pedigree completeness.

Supplementary Table S6. Published Ashkenazi Jewish carrier-frequency estimates for the *DHDDS* c.124A>G founder allele.

Reference	Year	Cohort size (n)	Carrier alleles observed	Estimated carrier freq.	Estimated population AF
Züchner et al. (AJHG)	2011	2,769	9	1 in 322 (0.31%)	$1.6\times10^{\blacksquare 3}$
Björkhem et al. (IOVS)	2013	1,418	5	1 in 354 (0.28%)	$1.4\times10^{\blacksquare 3}$
Hadassah screen (2014)	2014	7,738	24	1 in 322 (0.31%)	$1.55\times10^{\blacksquare 3}$
gnomAD v2.0.2 (AJ subset)	2016	5,076	16	1 in 317 (0.32%)	$1.58\times10^{\blacksquare 3}$

Carrier alleles observed refers to heterozygous individuals identified during population screening, NOT to homozygous proband alleles. These figures are provided for context and are not part of the causal-variant tally reported in the main manuscript.